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Proof of Concept Strategy for TransCelerate DDF Commercial Adaptation

1. POC Objectives

- **Validate Technical Feasibility**
 - Demonstrate successful deployment of core SDR components
 - Verify integration with at least one upstream and one downstream system
 - Confirm data model compatibility and transformation capabilities
- **Demonstrate Business Value**
 - Measure time savings in study setup processes
 - Quantify reduction in manual configuration efforts

- Assess data quality improvements
- **Identify Implementation Challenges**
 - Uncover integration complexities
 - Assess performance and scalability
 - Identify organizational change management needs
- **Refine Implementation Approach**
 - Validate or adjust cost and timeline estimates
 - Refine technical architecture
 - Develop detailed scaling strategy

2. POC Scope

In Scope:

- Core SDR deployment with essential components
- Integration with one study design system (upstream)
- Integration with one EDC system (downstream)
- Implementation of limited BiomedicalConcept set
- End-to-end workflow for a simplified study design
- Basic metrics collection and analysis

Out of Scope:

- Full production deployment
- Complete system integrations
- Comprehensive BiomedicalConcept library
- Advanced analytics capabilities
- Organization-wide process changes

3. POC Technical Architecture

Component Selection

1. **Study Definition Repository (SDR)**

- Deploy core API components from TransCelerate repository
- Configure MongoDB/CosmosDB for data storage
- Implement basic authentication
- Enable USDM version 3.0 support

2. System Adapters

- Develop adapter for selected study builder (upstream)
- Implement adapter for target EDC system (downstream)
- Focus on core transformation capabilities

3. Metrics Collection

- Implement logging for key operations
- Track timing for critical processes
- Capture error rates and types
- Monitor system performance

4. Administration Interface

- Basic configuration management
- System status monitoring
- User management
- BiomedicalConcept management

Technology Stack

Component	Technology	Justification
Backend API	.NET 6	Matches TransCelerate implementation
Database	MongoDB/CosmosDB	Document storage for USDM JSON
Adapters	.NET 6 or Node.js	Flexibility for system integration
Admin UI	Angular	Matches TransCelerate UI implementation
Metrics	Application Insights	Cloud-native monitoring
Deployment	Azure App Service	Simplified deployment and scaling

4. POC Implementation Plan

Phase 1: Setup and Configuration (Weeks 1-2)

Key Activities:

- Set up development environment
- Deploy core SDR components
- Configure database
- Implement basic authentication
- Establish metrics collection

Deliverables:

- Functioning SDR instance
- API documentation
- Environment configuration guide
- Initial metrics dashboard

Phase 2: Upstream Integration (Weeks 3-4)

Key Activities:

- Select study builder system
- Develop integration adapter
- Implement USDM transformation
- Test with sample study designs
- Measure performance and accuracy

Deliverables:

- Working upstream adapter
- Integration documentation
- Test results report
- Performance metrics

Phase 3: Downstream Integration (Weeks 5-6)

Key Activities:

- Select EDC system

- Develop EDC adapter
- Implement form generation
- Test end-to-end workflow
- Measure time savings and quality

Deliverables:

- Working downstream adapter
- Integration documentation
- End-to-end test results
- Comparative metrics report

Phase 4: Evaluation and Planning (Weeks 7-8)

Key Activities:

- Conduct comprehensive testing
- Analyze metrics and results
- Document challenges and solutions
- Develop scaling strategy
- Prepare final report and recommendations

Deliverables:

- POC evaluation report
- Implementation recommendations
- Scaling strategy
- Executive presentation

5. BiomedicalConcept Selection for POC

For the POC, we need a representative but manageable set of BiomedicalConcepts:

Core BiomedicalConcept Set

1. Vital Signs Category

- Blood Pressure (systolic, diastolic)

- Heart Rate
- Respiratory Rate
- Temperature
- Height/Weight/BMI

2. Laboratory Category

- Complete Blood Count (basic parameters)
- Basic Metabolic Panel (key electrolytes)
- Liver Function Tests (basic parameters)

3. Patient Reported Outcomes

- Pain Assessment (numeric scale)
- Quality of Life (simple questionnaire)

4. Safety Assessments

- Adverse Event Recording
- Concomitant Medication

This set provides:

- Mix of numeric, coded, and text data
- Common measurements across studies
- Representation of different assessment types
- Manageable scope for POC

6. Metrics and Success Criteria

Technical Metrics

Metric	Measurement Method	Success Threshold
API Response Time	Direct measurement	90% of requests < 500ms
Transformation Accuracy	Validation against reference	> 99% field accuracy
System Uptime	Monitoring	> 99% during POC
Error Rate	Log analysis	< 1% of operations

Business Value Metrics

Metric	Measurement Method	Success Threshold
Study Setup Time	Comparative timing	> 50% reduction
Manual Configuration Effort	Task timing	> 70% reduction
Form Generation Accuracy	Manual validation	> 95% accuracy
Edit Check Generation	Comparative analysis	> 80% auto-generated

7. Risk Management

Identifying and mitigating risks is essential for POC success:

Risk	Probability	Impact	Mitigation Strategy
Integration technical challenges	High	Medium	Start with simpler systems, allow buffer time
Performance issues	Medium	Medium	Implement monitoring, optimize early
Data model compatibility	Medium	High	Start with core concepts, validate mappings
Resource constraints	Medium	High	Clear scope definition, phased approach
Stakeholder expectations	High	Medium	Clear communication, expectation management

8. Next Steps

To initiate the POC, we recommend these immediate actions:

- Secure Executive Sponsorship
 - Present POC plan to key stakeholders
 - Secure resource commitments

- Establish governance structure

2. Form POC Team

- Identify and assign team members
- Conduct kickoff meeting
- Establish working agreements

3. Set Up Development Environment

- Clone TransCelerate repositories
- Configure development infrastructure
- Establish CI/CD pipeline

4. Select Target Systems

- Identify study builder for upstream integration
- Select EDC system for downstream integration
- Document system interfaces and requirements

5. Define Test Study

- Create or select study design
- Define BiomedicalConcepts to include
- Establish reference implementation